

Arnav Pundir

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PROFESSIONAL SUMMARY

Computer Vision and AI-focused Software Engineer specializing in Face Recognition, Object Detection, Temporal Re-Identification, and Multimodal LLMs. Proven track record of engineering deployable real-world AI pipelines using YOLOv8, InsightFace ArcFace, OpenCV, and Gemini APIs integrated into high-performance Python backends (Flask/FastAPI) with vector databases.

TECHNICAL SKILLS

ML & Computer Vision : OpenCV, YOLOv8, InsightFace (ArcFace), ByteTrack, MediaPipe, PyTorch
NLP & AI Frameworks : Gemini 2.5 API, LLM Prompt Engineering, Vector Search, NumPy, Scikit-Learn
Languages & Backend : Python, C/C++, SQL, JavaScript, TypeScript, FastAPI, Flask, REST APIs, WebSockets
Databases & Cloud : Supabase (pgvector), SQLite, PostgreSQL, MongoDB, Azure AI Vision, Docker, Git

EDUCATION

COER University	Roorkee, India
<i>Bachelor of Technology in Computer Science and Engineering</i>	2024 – 2028
Atal Utkrisht Gov. Inter College	Roorkee, India
<i>Senior Secondary (Class XII) PCM</i>	2022 – 2024

PROFESSIONAL EXPERIENCE

Freelance Software Developer	Jan 2024 – Present
<i>Client & Independent Projects</i>	Roorkee, India
- Engineered scalable, high-performance web platforms (e.g., SevaSetu) using React.js and robust REST APIs, evaluating architectural trade-offs for complex client requirements. - Architected interactive WebGL applications utilizing Three.js, critically analyzing system performance bottlenecks to consistently achieve 95+ Lighthouse accessibility and speed scores.	

AI & COMPUTER VISION PROJECTS

BioSecure AI Facial Recognition & Template Aging Engine | [GitHub](#)

- Engineered a 512D ArcFace facial embedding pipeline via **InsightFace** paired with **Supabase pgvector** for instant cosine similarity matching, achieving **90%+ multi-face detection accuracy**.
- Developed a novel **3D Pose-Gated EWMA Embedding Drift Engine** ($\alpha = 0.30$) to monitor template aging and mitigate biometric decay before recognition failure occurs.
- Built a production-ready **Flask** web application with multi-tier SMTP alert escalation and automatic report generation, reducing manual roll call effort by **80%**.

TraceAI Forensic Surveillance Intelligence Platform | [GitHub](#)

- Developed a forensic surveillance platform orchestrating **YOLOv8n**, **ArcFace**, and **ByteTrack** to track and re-identify persons across up to 16 concurrent video feeds at **15 FPS**.
- Engineered spatial-temporal movement timeline reconstruction and dwell-time analytics across camera zones.
- Architected a **FastAPI** asynchronous backend pushing real-time watchlist hit notifications via **WebSockets**.

Maya AI Desktop Voice & Multimodal Assistant | [GitHub](#)

- Architected a multi-layer AI assistant utilizing **Gemini 2.5 Flash** with rule-based offline fallback for robust continuous execution.
- Built a real-time **PyQt5** GUI featuring reactive waveform visuals and multi-threaded audio input processing.
- Integrated OS-level automation for audio control, screen dimming, screenshot capture, and automated web parsing.

DocBot Intelligent NLP Document Summarizer | [GitHub](#)

- Architected an NLP application leveraging the **Gemini API** to autonomously parse web documentation and resolve complex user queries in real-time.
- Built a native **PyQt5** interface efficiently orchestrating live content extraction and semantic text processing.

CERTIFICATIONS & OPEN SOURCE

Core Contributor	Apr 2026 – Present
<i>Nexus Spring of Code (NSoC '26) & GirlScript Summer of Code (GSSoC)</i>	Global Rank #70
Achieved Global Rank #70 by merging 13 Pull Requests across 5 open-source repositories, implementing HttpOnly cookie authentication, tightening CORS security, and improving performance.	
Microsoft Learn & Google Certifications	2025 – 2026
<i>Azure AI Vision & Generative AI</i>	Online
Earned certifications in Object Detection, OCR, Face Recognition, and Foundational Large Language Models.	